RR-L-112b
OCTOBER 24, 1960
SUPERSEDING
Fed. Spec. RR-L-112
September 18, 1958

FEDERAL SPECIFICATION

LANTERNS, KEROSENE, HAND, GLOBE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers kerosene-burning hand lanterns with or without glass globes and also provides for the procuring of glass globes separately (see 6.1).

1.1.1 *Federal specification coverage.*—This specification does not include all types and classes of Lanterns, Kerosene, Hand, Globe, but is intened to cover only those types and classes generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* — The lanterns shall be of the following types and classes (see 6.1):

Type I. — Cold blast, tubular frame.
Class A. — Large size, medium globe (style F), 32-ounce capacity, 7/8-inch wick, 45 hours burning time.
Class B. — Small size, short globe (style B), 16-ounce capacity, 5/8-inch wick, 30 hours burning time.
Class C. — Small size, short globe (style B), 32-ounce capacity, 1 3/32-inch wick, 100 hours burning time.

Type II. — Cold blast, wire and tubular frame, short globe (style E), 5/8-pint capacity, 31/64-inch wick, 12 hours burning time.

Type III. — Hot blast, tubular frame, high globe (style A), 1-pint capacity, 5/8-inch wick, 30 hours burning time.

Type IV. — Wire-Frame, short globe (style C or D), 3/8-pint capacity, 5/8-inch wick, 20 hours burning time.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-C-311 — Chimmeys and Globes (Lamp and Lantern); Glass.
QQ-T-~~181~~ — Terne Plates (for Manufacturing Purposes).
QQ-T-425 — Tinplate (Hot Dip and Electrolytic).
QQ-W-461 — Wire, Steel, (Carbon), Round, Bare and Coated.
QQ-Z-325 — Zinc Coating, Electrodeposited, Requirements for.
UU-T-~~111~~ — Tape; Paper, Gummed (Sealing and Securing).
VV-K-211 — Kerosene.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 — Boxes, Wood, Cleated-Plywood.

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PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636 — Box, Fiberboard.
PPP–C–843 — Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification, to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois).

3. REQUIREMENTS

3.1 Preproduction sample.—Unless otherwise specified (see 6.1), before production is commenced a sample of the finished commodity shall be submitted or made ready for approval of the contracting officer or his authorized representative for inspection and tests as specified in section 4 to determine compliance with this specification. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification.

3.2 Materials. — The materials specified shall conform to the applicable specifications and requirements specified hereinafter.

3.2.1 *Steel, carbon.*

3.2.1.1 *Uncoated and coated sheet or strip.* — Uncoated sheet or strip, where specified herein, shall be either bright or black. Coated sheet steel, where specified herein, shall be either zinc-coated, tin-coated, or terne-metal coated. The thickness of the sheet, before any coating or finish is applied, shall be not less than 0.0121-inch to a maximum of 0.0149-inch nominal. The coating shall be tightly adherent and shall thoroughly and evenly cover the base sheet.

3.2.1.1.1 *Zinc-coating thickness.* — Zinc-coating thickness of each surface shall conform to the requirements of type II, class 2 of Federal Specification QQ–Z–325.

3.2.1.1.2 *Terne metal coating.* — Terne metal coating shall be grade A, 8 pounds per

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double base box, conforming to Federal Specification QQ–T–181.

3.2.1.1.3 *Tin coating thickness.* — The tin-metal coating shall conform to type I, grade 4, class 25 of Federal Specification QQ–T–425.

3.2.1.2 *Wire, steel.* — The round steel wire shall conform to grade FS 1010–1015 of Federal Specification QQ–W–461.

3.2.2 *Globe, glass.* — The glass globes shall be heat-resisting glass or commercial lime quality glass, clear or colored, as specified (see 6.1). If the kind of glass is not specified, the contractor may, at his option, furnish either heat-resisting glass or commercial lime quality glass globes.

3.2.2.1 *Heat-resisting glass globes.* —The globes shall conform to Federal Specification DD–C–311, type I, class A or class B.

3.2.2.2 *Commercial lime quality glass globes.* — The globes shall be of good quality, free from heavy cords, stones, and fire cracks. Scattered bubbles are permissible provided they do not interfere with the performance of the globes. Surfaces shall be polished sufficiently to eliminate dullness.

3.2.2.3 *Clear globes.* — Clear globes shall conform to type I, class A of Federal Specification DD–C–311.

3.2.2.4 *Colored globes.* — Colored globes shall be of solid-color glass conforming to type I, class B of Federal Specification DD–C–311.

3.2.3 *Wick.* — The wick shall be of a standard commercial grade of cotton woven material of the sizes specified hereinafter.

3.3 Design and construction (general).

3.3.1 *Design.*

3.3.1.1 *Leakage.* — The lantern shall be oiltight at the seams when tested as specified in 4.4.2.1. The burner and the filler hole shall show no leakage when tested as specified in 4.4.2.2.

3.3.1.2 *Fount.* — When the lantern is placed on a horizontal plane surface, the bottom of the fount shall not contact the plane except around the rim, and the lantern shall not rock more than $\frac{1}{16}$ inch nor lean more than 5 degrees.

3.3.1.3 *Burner.* — The burner shall be a detachable unit. The base shall snugly fit the receiving collar on the fount. The burner shall have a wick spindle having two securely fastened toothed wheels. The wick-raiser spindle shall be suitable for raising and lowering the wick from the outside of the lantern.

3.3.1.4 *Lock.* — The burner shall be mechanically locked into fount by means of a bayonet-type lock formed by the wick-raiser spindle and fount, or by a wing-type lock formed by the burner cap and fount.

3.3.1.5 *Illuminant.* — The burner shall be designed to use kerosene as the illuminant. The kerosene shall comply with the requirements of Federal Specification VV–K–211. The burner shall maintain a steady, smokeless flame when tested as specified in 4.4.6. Soot formation on the globe or excessive build-up of soot on the interior of the super structure which would impair the normal burning of the lantern shall be cause for rejection.

3.3.1.6 *Globe-lifting mechanism.* —When a globe-lifting mechanism is required, the globe shall not fall out when the lantern is held in any position and the globe-lifting mechanism is rapidly operated between the limits of the lifting and lowering motions.

3.3.1.7 *Filler hole and cap.* — With exception of the type II lantern, the filler-hole collar shall be coated sheet steel, threaded, and joined to the fount securely. The filler

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cap shall be coated sheet steel, or molded composition, threaded to engage the threads of the filler hole and provided with a fiber washer to make the filler hole oiltight.

3.3.1.8 *Bail.* — The normal position of the bail shall be such that it will remain sufficiently cool for handling when tested as specified in 4.4.3. With the bail upright, the distance between the top of the bail and the top of the dome shall be not less than four inches.

3.3.1.9 *Globe, style, and dimensions.* — For the style hereinafter specified, the shape of globe shall be in substantial agreement with the referenced figures, and the dimensions shall comply with those specified in table I. The globe shall be readily removable without use of tools, excessive pressures, or interference with assembled members.

3.4 Design and construction (details).

3.4.1 *Type I, cold blast-tubular frame.*

3.4.1.1 *Design.*—The type I, cold blast lantern shall have two hollow side tubes, which shall form air passages between the dome and the base of the burner support. Fresh air shall be drawn into the tubes at the top after having been warmed by the heated dome. The opening above the wick in the burner cap shall provide the outlet for the gases drawn into the base of the burner support through the side tubes. The lantern shall have a fount, burner, burner cap, two globe-guard wires, globe-lifting mechanism, filler hole and cap, bail, wick, dome, and globe.

3.4.1.2 *Classes.*

3.4.1.2.1 *Class A.* — The type I, class A lantern shall be in substantial agreement with figure 7, and shall conform to the dimensions given in table II. The globe shall be style A and the color shall be as specified (see 6.1). The capacity of the fount shall be not less than 32 fluid ounces. The lantern does not comply with the requirements for capacity if there is any overflow either above the fill-opening or into the burner-receiving collar when tested in accordance with 4.4.1. The burning time shall be not less than 45 hours when tested as specified in 4.4.6. The wick shall be $\frac{7}{8}$ inch wide (plus or minus $\frac{1}{16}$ inch) by $\frac{1}{16}$ inch thick nominal and not less than 7 inches long.

3.4.1.2.2 *Class B.* — The type I, class B lantern shall be in substantial agreement with figure 8 and shall conform to the dimensions given in table II. The globe shall be style B, and the color shall be as specified (see 6.1). The capacity of the fount shall be not less than 16 fluid ounces. The lantern does not comply with the requirements for capacity if there is any overflow either above the fill-opening or into the burner-receiving collar when tested in accordance with 4.4.1. The burning time shall be not less than 30 hours when tested as specified in 4.4.5. The wick shall be $\frac{5}{8}$ inch wide (plus or minus $\frac{1}{16}$ inch) by $\frac{1}{16}$ inch thick nominal and not less than 7 inches long.

3.4.1.2.3 *Class C.* — The type I, class C lantern shall be in substantial agreement with figure 8, and shall conform to the dimensions given in table II. The globe shall be style B, and the color shall be as specified (see 6.1). The capacity of the fount shall be not less than 32 fluid ounces. The lantern does not comply with the requirements for capacity if there is any overflow either above the fill-opening or into the burner-receiving collar when tested in accordance with 4.4.1. The burning time shall be not less than 100 hours when tested as specified in 4.4.5. The wick shall be $1\frac{3}{32}$ inch wide (plus or minus $\frac{1}{16}$ inch) by $\frac{1}{16}$ inch thick nominal and not less than 7 inches long.

3.4.1.3 *Side tubes.* — The side tubes shall form rigid connecting members between the dome and the fount. Each side tube shall be formed from two pieces of either uncoated or coated sheet steel which are joined together with two inserted joints and have two continuous flat clinched seams. Tube walls shall be either smooth or beaded. The outside diameter of the tubes shall be not less than

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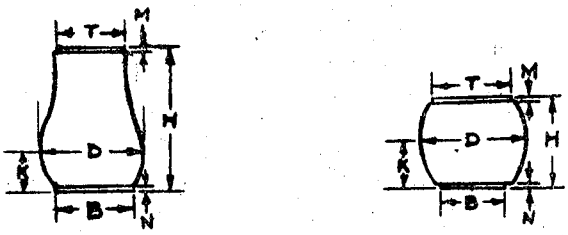


FIGURE 1.—Style A globe for type III lantern.

FIGURE 2.—Style B globe for type I, classes B and C lanterns.

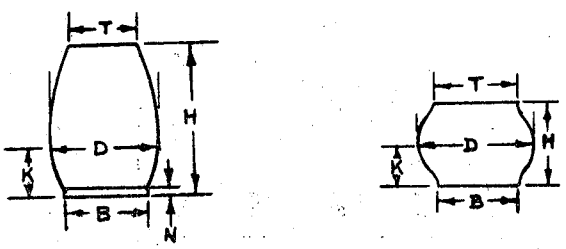


FIGURE 3.—Style C globe for type IV lantern.

FIGURE 4.—Style D globe for type IV lantern.

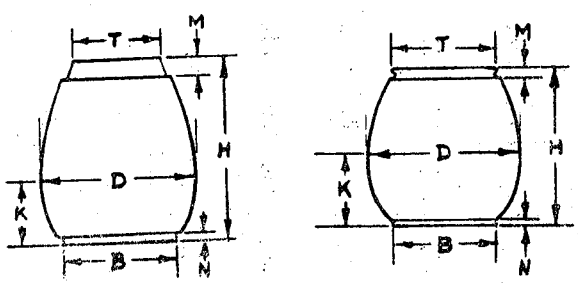


FIGURE 5.—Style E globe for type II lantern.

FIGURE 6.—Style F globe for type I, class A lantern.

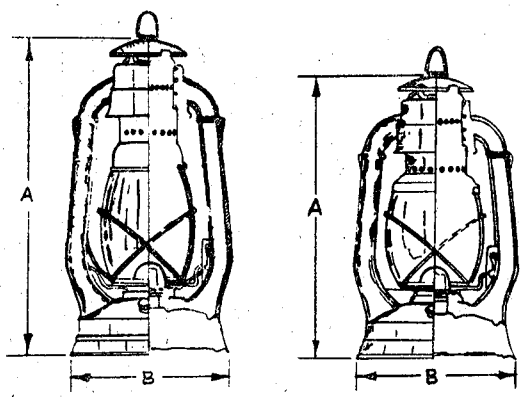


FIGURE 7.—Type I, class A, large size, medium globe. (See table II for dimensions A and B.)

FIGURE 8.—Type I, classes B and C, small size, short globe lanterns. (See table II for dimensions A and B.)

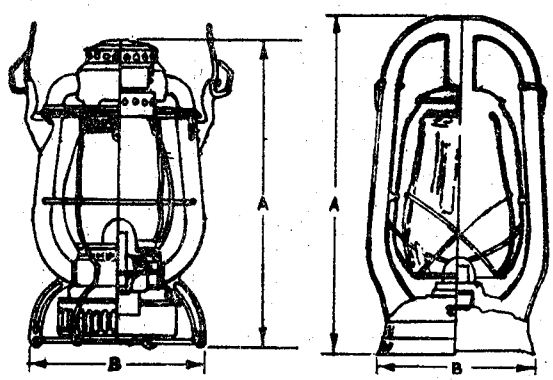


FIGURE 9.—Type II lantern. (See table II for dimensions A and B.)

FIGURE 10.—Type III lantern. (See table II for dimensions A and B.)

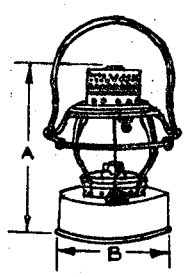


FIGURE 11.—Type IV lantern. (See table II for dimensions A and B.)

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TABLE I.—*Globes: dimensions*

	Outside rim diameter at bottom B			Outside rim diameter at top T			Over-all height H			Rim height at top M			Rim height at bottom N			Major diameter				
																Outside diameter D			Height above base K	
	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum	Minimum	Maximum
	Inch	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
Style A (see figure 1)	3 1/16	3 3/8	3 1/2	2 13/16	2 3/4	2 7/8	6 5/8	6 1/2	6 3/4	3/16	3/32	9/32	3/16	5/32	1/4	4 5/8	4 9/16	4 15/16	1 5/8	2
Style B (see figure 2)	2 13/16	2 3/4	2 7/8	3 1/4	3 3/32	3 3/8	3 13/16	3 3/4	4	3/16	3/32	9/32	3/16	5/32	1/4	4 1/8	3 15/16	4 1/4	1 1/2	1 3/4
Style C (see figure 3)	3 1/4	3 3/16	3 5/16	2 1/2	2 7/16	2 9/16	4 29/32	4 1/2	5 7/16	--	--	--	1/8	3/32	3/16	4 9/16	4 1/2	4 5/8	1 1/4	1 3/4
Style D (see figure 4)	3 1/4	3 3/16	3 5/16	3 1/4	3 3/16	3 5/16	3 1/4	3 3/16	3 5/16	--	--	--	--	--	--	4 15/32	4 13/32	4 17/32	1 1/2	1 3/4
Style E (see figure 5)	2 23/32	2 13/32	2 27/32	2 5/16	2 1/4	2 3/8	4 1/4	4 3/16	4 5/16	1/2	7/16	9/16	1/8	3/32	3/16	3 13/16	3 3/4	4 29/32	1 1/4	1 3/4
Style F (see figure 6)	3 1/16	3 3/8	3 1/2	3 1/2	3 1/16	3 9/16	4 5/8	4 9/16	4 11/16	1/4	3/16	5/16	1/8	3/32	3/16	4 3/8	4 5/16	4 7/16	2 1/8	2 1/4

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TABLE II.—*Lanterns, dimensions*

Type and class	Over-all height, bail down, plus or minus 10 percent A	Outside diameter of base, minimum B
	Inches	*Inches*
Type I:		
Class A (see figure 7)	13¼	6¾
Class B (see figure 8)	11½	5¾
Class C (see figure 8)	12	6¼
Type II (see figure 9)	9¾	6⅛
Type III (see figure 10)	13½	5¾
Type IV (see figure 11)	8¼	6

⅝ inch. The lower end of the tubes shall be joined to the fount or burner support with the seams completely soldered or welded. The tops of the tubes shall be inserted into the outer cylinder of the dome and shall be permanently secured.

3.4.1.4 *Dome.* — The dome shall consist of an outer cylinder, inner cylinder, inner baffle, helical wire spring, top plate, and wire pull. The bottom of the inner cylinder shall completely enclose the rim of the top of the globe. The top of the inner cylinder shall have openings to permit burned gases to escape. The inner cylinder shall extend upward through the outer cylinder and by means of the spring shall exert sufficient pressure to firmly and securely hold the globe in position. The top plate shall protect the cylinder openings against entrance of rain drops. The wire pull, when operated, shall permit releases of the pressure of the inner cylinder on the globe rim. The arrangement of inner baffle, and outer and inner cylinders shall be such as to make the normal flame of the wick windproof, and prevent mixing of burned gases with the fresh air drawn into the side tubes.

3.4.1.5 *Fount.* — The fount shall be coated sheet steel conforming to the requirements specified in 3.2.1. The fount bottom shall be joined to the body with an oiltight double-folded joint. The bottom shall be stiffened by means of concentric offsets, concaving or embossing.

3.4.1.6 *Burner cap.* — The burner shall be provided with a coated sheet steel or brass cap having an opening above the wick.

3.4.1.7 *Globe-guard wires.* — The globe-guard wires shall be zinc-coated and of a 0.105-inch-nominal diameter. The wires shall cross each other at diametrically opposite points adjacent to the unprotected surface of the globe and approximately two inches above the bottom edge of the globe. The wires shall be securely fastened to a coated sheet globe-supporting plate, and shall contact the globe at the points where they cross each other. The globe-supporting plate shall be perforated to admit fresh cold air into the globe below the flame.

3.4.1.8 *Globe-lifting mechanism.* — The globe-lifting mechanism shall consist of a formed wire lift with integral lever and a lever guide which locks the lever at the top and bottom of the lift positions. The wire lift shall be steel, zinc-coated, and of a 0.140-inch-nominal diameter. The wire lift shall be hinged to the globe-supporting plate and supported at both side tubes. The lever guide shall be securely welded or soldered to one side tube. The lever shall lift the assembly

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comprising inner cylinder, globe guards, and globe-supporting plate so the wick may be readily ignited with a match.

3.4.1.9 *Bail.* — The bail shall be formed of zinc-coated steel wire not less than 0.134-inch nor more than 0.156-inch diameter. The bail shall not be detachable when tested as specified in 4.4.4.

3.4.1.10 *Finish.* — Unless otherwise specified (see 6.1) the lantern shall be finished as follows, according to the kind of sheet metal used:

(*a*) The lantern shall have the natural zinc or terne-metal finish when the side tubes, the exterior members of the dome, and top and sides of the fount are zinc-coated, or terne-metal coated.

(*b*) The lantern shall have a durable, tightly adherent enamel, lacquer, or paint finish of dark green color when any side tube or exterior member of the dome is of uncoated or electrolytic tin-coated sheet steel.

3.4.2 *Type II, cold blast, wire frame, short globe (style E), 5/8-pint capacity 31/64-inch wick.*

3.4.2.1 *Design.* — The type II, cold blast, wire-frame lantern, shall have two hollow side tubes, which shall form air passages between the dome and the base of the burner support. Fresh air shall be drawn into the tubes at the top after being warmed by the heated dome. The perforated opening in the globe seat shall provide the outlet for the gases drawn into the base of the burner support through the side tubes. It shall have a wire frame made of 4 uprights, either straight or curved, and 2 horizontal wires. The 2 horizontal and 4 upright wires shall be 0.135-inch-nominal diameter. Each of the upright wires shall be securely interlocked with each of the horizontal wires, either by means of welded joints or formed wire loops. The frame shall be securely welded to the globe holder and fount holding member. The lantern shall have a dome, burner, bail, wick, and globe. The arrangement and fabrication of the dome and air intake member shall be such that the normal flame is unaffected by wind or rain. The lantern shall be in substantial agreement with figure 9, and shall conform to the dimensions given in table II. The globe shall be style E. Unless otherwise specified (see 6.1), a clear globe shall be furnished. The capacity of the fount, without any overflow into the burner receiving collar, shall be not less than 10 fluid ounces nor more than 14 fluid ounces. The wick shall be 31/64 inch wide (plus or minus 1/16 inch) by 1/16 inch thick nominal and not less than 7 inches long. The burning time shall be not less than 12 hours when tested as specified in 4.4.6.

3.4.2.2 *Dome.* — The dome shall consist of a perforated globe cover, baffles, globe-cushioning plate with wire spring, steel hinge, and a wire cover-locking clip. The cover, baffles, and cushioning plate shall be made of coated sheet steel. The dome shall be hinged to the frame and securely locked with the frame by means of the locking clip. The locking clip shall not release the cover when any opening force is applied with the fingers to the edge or any portion of the cover. The cushioning plate shall, by means of the spring, apply sufficient pressure to the top rim of the globe to firmly hold it in position.

3.4.2.3 *Fount.* — The fount shall be coated sheet steel conforming to the requirements specified in 3.2.1. The fount bottom shall be joined to the body with an oil-tight double-folded joint and stiffened by means of a reinforcing wire ring, concentric offsets, concaving, or embossing.

3.4.2.4 *Bail.* — The bail shall be formed of zinc-coated steel wire not less than 0.162 inch in diameter. The ends of the bail shall be looped and permanently attached to the frame.

3.4.2.5 *Finish.* — Unless otherwise specified (see 6.1) the lantern shall be finished as follows:

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(*a*) The lantern shall have the natural-zinc or terne-metal finish when the dome globe cover, globe cushioning plate, hinge, fount top, and fount body are zinc-coated, or terne-metal coated.

(*b*) The lantern shall have a durable, tightly adherent enamel, lacquer, or paint finish of dark green color when any dome globe cover, globe-cushioning plate, hinge, fount top, or fount body has a terne-metal coating lighter than 0.29 ounce per square foot.

3.4.3 *Type III, hot blast, high globe (style A) 1-pint capacity, 5/8-inch wick.*

3.4.3.1 *Design.* — The type III hot-blast lantern shall have two hollow side tubes, which shall form air passages between a central tube of the dome and the base of the burner support. A mixture of cold fresh air, and heated exhaust air shall be directed into the central tube by an inverted funnel projecting beyond the periphery of the globe top and providing a clear opening approximately 3/8 inch high around the top edge of the globe. The lantern shall have a fount, burner, burner cap, two globe-guard wires, globe-lifting mechanism, fill opening and cap, bail, globe-supporting wires, wick, and globe. The lantern shall be in substantial agreement with figure 10, and shall conform to the dimensions given in table II. The globe shall be style A, and the color shall be as specified (see 6.1). The capacity of the fount shall be not less than 16 fluid ounces nor more than 20 fluid ounces. The lantern does not comply with the requirements for capacity if there is any overflow either above the fill-opening or into the burner receiving collar when tested in accordance with 4.4.1. The burning time shall be not less than 30 hours, when tested as specified in 4.4.5. The wick shall be 5/8 inch wide (plus or minus 1/16 inch).

3.4.3.2 *Side tubes.* — The side tubes shall comply with the requirements of 3.4.1.3. except that the tops of the tubes shall be permanently secured to the outlets of the central tube.

3.4.3.3 *Dome.* — The dome shall consist of a central tube and an inverted funnel, both fabricated from either uncoated or coated sheet steel conforming to the requirements specified in 3.2.1. The central tube shall have an outside diameter of not less than one inch. The funnel shall gather and direct a portion of the rising hot gases into the central tube.

3.4.3.4 *Fount.* — The fount shall comply with the requirements of 3.4.1.5.

3.4.3.5 *Burner cap.* — The burner cap shall comply with the requirements of 3.4.1.6.

3.4.3.6 *Globe-guard wires.* — The globe-guard wires shall comply with the requirements of 3.4.1.7.

3.4.3.7 *Globe-lifting mechanism.* — The globe-lifting mechanism shall comply with the requirements of 3.4.1.8 except that the lever shall lift the assembly comprising the funnel, globe, globe guards, globe supporting plate and wires.

3.4.3.8 *Bail.* — The bail shall comply with the requirements of 3.4.1.9.

3.4.3.9 *Globe-supporting wires.* — The globe-supporting wires shall consist of 2 upright wires and a globe-rim-holding wire. The wires shall be steel, zinc-coated, and of a 0.105-inch nominal diameter. The upright wires shall be securely fastened to the funnel and the globe-supporting plate at the top and bottom, respectively. Either the upright wires shall be arranged to be in contact with and press against the globe to hold it securely, or the globe guard wires shall support the globe at the sides adjacent to the tubes. The rim-holding wire shall encircle the projecting rim of the globe top and shall be so arranged that by lifting or pressing a formed wire catch, the pressure on the rim may be relieved and the globe readily removed from the lantern.

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3.4.3.10 *Finish.* — Unless otherwise specified, the lantern shall be finished as specified in 3.4.1.10.

3.4.4 *Type IV, wire frame, short globe (style C or D), ⅜-pint capacity, ⅝-inch wick.*

3.4.4.1 *Design.* — The type IV, wire-frame lantern shall have a frame made of 6 or more upright supports, either straight or curved, and 2 or more horizontal wires. The horizontal wires and at least 4 of the upright wires shall be not less than 0.148 inch in diameter. Each of the upright wires shall be securely interlocked with each of the horizontal wires, either by means of welded joints or formed wire loops. The frame shall be securely welded to either the fount or air intake member. The lantern shall have a dome, burner, fill opening and cap, bail, wick, and globe. The arrangement and fabrication of the dome and air intake member shall be such that the normal flame is unaffected by the wind or rain. The lantern shall be in substantial agreement with figure 11, and shall conform to the dimensions given in table II. The globe shall be either style C or D at the option of the contractor. Unless otherwise specifiied (see 6.1), a ruby globe shall be furnished. The capacity of the fount, without any overflow into the burner receiving collar, shall be not less than 6 fluid ounces, nor more than 16 fluid ounces. The wick shall be ⅝ inch wide (plus or minus 1/16 inch) by 1/16 inch thick and not less than 7 inches long. The burning time shall be not less than 20 hours when tested as specified in 4.4.5.

3.4.4.2 *Dome.* — The dome shall comply with the requirements of 3.4.2.2.

3.4.4.3 *Fount.* — The fount shall be coated sheet steel conforming to the requirements specified in 3.2.1. The fount bottom shall be joined to the body with an oil-tight double-folded joint, and shall be stiffened by means of a reinforcing wire ring, concentric offsets, concaving, or embossing.

3.4.4.4 *Air-intake.* — The air-intake member shall be either uncoated or coated sheet steel and permanently secured to the fount with an oiltight joint. It shall have perforations through which air is supplied to maintain the burner flame. It shall have a burner-receiving collar, of either uncoated or coated sheet steel which shall be either soldered to the fount, or permanently fastened to the fount with an oiltight inserted joint.

3.4.4.5 *Bail.* — The bail shall be formed of zinc-coated steel wire not less than 0.134 inch nor more than 0.156 inch in diameter. The ends of the bail shall be looped and permanently attached to the frame.

3.4.4.6 *Finish.*—Unless otherwise specified (see 6.1) the lantern shall have a durable, tightly adherent red-enamel finish free of cracks, blisters or other film imperfections.

3.5 Identification marking. — The lantern shall be permanently embossed with the letters "U.S.", manufacturer's model designation, the model number if any, and the manufacturer's name or trademark of such known character as to be easily identifiable with said manufacturer. Letters, numerals, and other characters shall be of such size as to be clearly legible.

3.6 Workmanship.—The lantern shall be clean, well finished, and shall be free from any defects which may affect serviceability or appearance.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility.—The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government re-

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serves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. — When required, examination and testing of the preproduction sample shall be made of a completely fabricated item for all provisions of this specification applicable to end product examination and tests.

4.3 Sampling for inspection and acceptance. — Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL–STD–105, except where otherwise indicated hereinafter.

4.3.1 *Inspection of materials and components.* — The supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specification and standards to the extent specified herein, or if none in accordance with this specification.

4.3.2 *Inspection of the end item.*

4.3.2.1 *Examination of the end item.* — Examination of the lantern shall be in accordance with the classification of defects and acceptable quality levels (AQL's) set forth as follows. The lot size shall be expressed in units of lanterns for the purpose of determining the sample size in accordance with Military Standard MIL–STD–105. The sample unit for this examination shall be one completely fabricated lantern.

4.3.2.1.1 *Examination for visual characteristics.*

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Classification of defects

Examine	Defects	Classification	
		Major	Minor
Finish of plated surfaces	Not specified type, evidence of corrosion; area of no coating or coating not adherent, blistered or peeling	X	
	Stained, discolored, spotted, gritty surface or otherwise not smooth		X
Finish of enamel, lacquer or painted surfaces	Not specified type or area of no coating	X	
	Cracked, peeling, lifted, blistered or gritty surface or not smooth		X
Quality of metal components	Not material specified or material broken, cracked or fractured	X	
	Dented, damaged bent, malformed affecting appearance		X
	Fin, burr or sharp edge which might cause injury	X	
Quality of glass components	Cracked, broken or not color specified	X	
	Bubbles affecting performance	X	
	Not polished as specified		X
	Heavy stones, cords or fine cracks		X
Soldering and welding	Missing, broken, not fused or metal burned through	X	
	Incomplete or splatter or flux not removed or not neat		X
	Solder not adherent	X	
Construction, assembly and workmanship	Not constructed as specified	X	
	Component not formed or attached as specified		X
	Component missing	X	
	Lantern leans more than 5 degrees when placed on level surface		X
	Lantern rocks more than 1/16 inch when placed on level surface	X	
	Fill opening not below lowest level of the burner receiving collar (except type II lantern)	X	
	Fill opening collar or cap not threaded	X	
	Foreign matter in fount	X	
	Globe not firmly or securely held in position. (When in normal use position)	X	
	Globe not readily removable	X	
	Wick raiser spindle does not have two toothed wheels		X
	Wick cannot be raised or lowered from outside of lantern	X	
	Burner unit not detachable	X	
	Burner not mechanically locked		X
	Wire pull at dome does not operate as specified (type I lanterns only)		X
	Wire cover locking clip dome does not operate as specified (types II and IV lanterns only)		X
Functioning of moving parts	Globe falls out when globe-lifting mechanism is rapidly operated between high and low limits	X	
	Evidence that wick cannot be readily ignited with a match when globe-lifting mechanism is in open position	X	
Identification marking	Missing, incorrect, illegible, incomplete or not embossed		X

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4.3.2.1.2 *Dimensional examination.* — Inspection will be made of lanterns for compliance with dimensions specified. Any deviation therefrom shall constitute a defect.

4.3.2.1.3 *Examination of preparation for delivery.* — An examination will be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be scored as specified below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot.

Examination	Defects
Markings, exterior and interior	Incorrect; illegible; omitted; of improper size, location, sequence, or method or application.
Materials	Any non conforming component; component missing, damaged, or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling; bulging or distortion of containers.
Content of container	Number per container is more or less than required.

4.3.2.1.4 *Inspection levels and AQL's.* — The acceptable quality levels (AQL's) expressed in defects per 100 units, and inspection levels shall be as follows:

	Inspection level	AQL's	
		Major	Total
For examination in 4.3.2.1.1	II	1.5	6.5
For examination in 4.3.2.1.2	L6	One-class	10.0
For examination in 4.3.2.1.3	L4	One-class	4.0

4.3.2.1.5 *Testing of the end item.*—Testing shall be performed in accordance with table III for the characteristics specified therein. The inspection levels applicable to each lot or batch shall be as indicated in table III. The sample unit for testing shall be one lantern with requirements applicable to individual units.

4.4 Tests.

4.4.1 *Capacity tests.* — Using an adequate liquid measure, the specified amount of kerosene shall be poured into the fount in accordance with the requirements specified in 3.4.1.2.1, 3.4.1.2.2, 3.4.1.2.3, 3.4.2.1, 3.4.3.1, and 3.4.4.1.

4.4.2 *Leakage.*

4.4.2.1 *Seams.* — The lantern, filled to capacity with kerosene shall stand on a level surface for 1 hour. If kerosene escapes, the seams are not oiltight.

4.4.2.2 *Burner and fill opening.* —The lantern, filled to capacity with kerosene and not ignited, shall be inclined at an angle of

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TABLE III.

Characteristic	Requirement paragraph	Test procedure	Inspection level	AQL
Capacity	3.4.1.2.1, 3.4.1.2.2, 3.4.1.2.3, 3.4.2.1, 3.4.3.1 and 3.4.4.1 as applicable	4.4.1	1	---
Leakage at seams, burner and fill opening	3.3.1.1, 3.4.1.5, 3.4.2.3, and 3.4.3.4 as applicable	4.4.2.1 and 4.4.2.2	L2	4.0
Heat transmission	3.3.1.8	4.4.4	1	---
Pull of bail	3.4.1.9	4.4.5	L2	4.0
Burning time	3.4.1.2.1, 3.4.1.2.2, 3.4.1.2.3, 3.4.2.1, 3.4.3.1 and 3.4.4.1 as applicable	4.4.6	1	---
Smokeless flame	3.3.1.5	4.4.7	1	---

[1] Tests for these characteristics shall be conducted on one sample lantern from initial production and need not be repeated unless there is a change in design, material or workmanship.

45 degrees with the vertical, preferably by securing its base to a surface which is inclined 45 degrees with the horizontal. If kerosene escapes at either the burner receiving collar or fill opening, the lantern is not oiltight. During the test, the filler cap shall be secured and fill opening shall be at the lowest elevation.

4.4.3 *Heat transmission.* — After the lantern has been burning with a normal flame for three hours, the lantern shall be picked up by the bail and checked for conformance to the requirements specified in 3.3.1.8.

4.4.4 *Pull of bail.* — A pull of 30 pounds shall be applied at either end of the bail, normal to the side tubes, with the bail either down or upright.

4.4.5 *Burning time.* — With the lantern filled to capacity and the flame adjusted to normal size the burning time shall be tested without any further adjustment or addition of kerosene.

4.4.6 *Smokeless flame.* — The lantern, filled to capacity and with the wick ignited, shall burn with a normal flame for three hours and shall conform to the requirements specified in 3.3.1.5.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment (for civil agencies only.)

5.1 Packaging.—Packaging shall be level A or C as specified (see 6.1).

5.1.1 *Level A.*

5.1.1.1 *Lantern, complete with globe.* — The globe of each lantern shall be wrapped in cellulosic cushioning material conforming to type III, class A of Federal Specification PPP–C–843. Sufficient cushioning material shall be used over the top and bottom edges

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of the globe to prevent movement or vibration while in place within the lantern frame.

5.1.1.2 *Lantern, without globe (when procured as a separate item) (see 6.1).*—Lanterns without globes shall not require packaging.

5.1.1.3 *Globes only (when procured as a separate item) (see 6.1).* — Globes shall be completely wrapped with cushioning material specified in 5.1.1.1. The wrap shall be secured with 1-inch width gummed paper tape conforming to class 2 of Federal Specification UU–T–111.

5.1.2 *Level C.* — Lanterns with globes, lanterns without globes, and globes only shall be packaged in accordance with the manufacturer's commercial practice.

5.2 **Packing.**—Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 *Level A.* — Twelve lanterns complete, or twelve lanterns without globes, or twelve globes of one type and class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC–L of Federal Specification PPP–B–636. Alternatively, shipping containers may be used that conform to overseas type of Federal Specification PPP–B–591, PPP–B–601, or Military MIL–B–10377; class 2, style 2 or 4 of Federal Specification PPP–B–621; or style 1, 2, or 2A, class 3 use, type 2 load of Federal Specification PPP–B–585. Each shipping container shall be provided with top and bottom pads and full height interlocking cell spacer made from B-flute fiberboard and a taped one-piece double-walled liner made from AB-flute fiberboard having a minimum bursting strength of 275 p.s.i. The B-flute of the liner shall be next to the item being packed. Each shipping container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to Military Specification MIL–L–10547. The liner shall be placed between the double-walled corrugated liner and the inner surfaces of the shipping container. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 *Level B.* — Twelve lanterns complete, or twelve lanterns without globes, or twelve globes of one type and class only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of Federal Specification PPP–B–636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Alternatively, shipping containers may be used that conform to domestic type, style A or B of Federal Specification PPP–B–591, PPP–B–601, or Military MIL–B–10377; class 1, style 2 or 4 of Federal Specification PPP–B–621; or style 1, 2, or 2A, class 1 use, type 2 load of Federal Specification PPP–B–585. Each shipping container shall be provided with top and bottom pads and full height interlocking cell spacers made from B-flute fiberboard and a taped one-piece double-walled liner made from AB-flute fiberboard having a minimum bursting strength of 275 p.s.i. The B-flute of the liner shall be next to the item being packed.

5.2.3 *Level C.* — Lanterns with globes, lanterns without globes, and globes only, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agency requirements.* — In addition to any special marking required by the contract or order, interior packages and

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shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military requirements.* — In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2.1 *Special marking.* — In addition to markings specified, the top of each shipping container containing lanterns complete or globes only shall be marked in characters not less than one inch in height as follows:

FRAGILE — GLASS
HANDLE WITH CARE

6. NOTES

6.1 Ordering data. — Purchasers should excercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Whether lanterns are to be procured without globes.
(*c*) Whether globes are to be procured as separate items.
(*d*) Type of lantern, and for type I lanterns the class also (see 1.2.1).
(*e*) Whether or not requirement of preproduction sample is waived (see 3.1).
(*f*) Whether heat resisting glass or commercial lime quality glass is required (see 3.2.2).
(*g*) The color of the globe (clear, blue, green, ruby, yellow, et cetera) (see 3.4.1.2.1, 3.4.1.2.2, 3.4.1.2.3, 3.4.2.1, 3.4.3.1 and 3.4.4.1.)
(*h*) Type of finish required if other than specified (see 3.4.1.10, 3.4.2.5, 3.4.3.10 and 3.4.4.6).
(*i*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Transportation description.—Transportation description applicable to this item is:

Lantern:

Lanterns, metal, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 20,000 pounds.

Globes:

Lantern globes, other than mica.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 20,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherother person or corporation, or conveying any rights or permission to manufacture, use, or sell any wise as in any manner licensing the holder or any patented invention that may in any way be related thereto.

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